

Question	Marking Guidance	Mark	Comments
04.1	1. Enters by diffusion; 2. Down a concentration gradient OR From high to low pO₂; 3. More/most across parts of body with gills; 4. Gills provide a larger surface area (for absorption); 5. 8.8 (kPa) over gills; 6. 2.4 (kPa - rest of body surface) / 1.9 (kPa - front end before gills) / 0.5 (kPa - rear end after gills);	4 max (4 x AO3)	1. Reject facilitated diffusion 1 and 2 'down a diffusion gradient' = 2 marks 2. Reject 'along' for 'down' 2. Accept description of O₂ is always higher in the water than the lugworm 4. Accept Gills increase SA:volume ratio

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04.2	Correct answer for 3 marks, 9.3×10^{-6} / 0.000 0093;;; MP1 – correct reading from graph (1.5) MP2 – correct rearrangement of equation ($\text{CdO}_2 = 0.000\ 031 \times \text{their pO}_2$) MP3 – their $\text{CdO}_2 \times 0.2$ OR their $\text{CdO}_2 \div 5$	3 (3 x AO2)	Accept correct rounding of 9.3×10^{-6}

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04.3	1. (Measure light) absorption/transmission; 2. Interpolate/draw line to curve/line then to pO_2 OR Read off (pO_2 figure) against absorbance/transmission value obtained;	2 (2 x AO3)	1. Accept 'absorbance' for absorption